Area and Perimeter Grid

TEACHER KEY

3rd Grade Math · Interactive Math · Explore & Reflect

Get started

Build rectangles on a grid or paint your own shapes from unit squares. Area and perimeter update as you go, so you can find shapes with the same area and different perimeters. Open the tool and use it to answer each question.

Explore

1. What are the area and perimeter of a rectangle 6 units long and 4 units wide?

Area 24 square units; perimeter 20 units

2. Find two rectangles with an area of 18 square units.

For example 3 by 6 and 2 by 9 (also 1 by 18 on a big enough grid)

3. What is the perimeter of a 5 by 5 square?

20 units

4. Rectangles with perimeter 12 can be 1 by 5, 2 by 4 or 3 by 3. What is the area of each?

5, 8 and 9 square units

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Explore more

5. A rectangle has a perimeter of 14 units and a width of 3 units. How long is it?

4 units (3 + 4 + 3 + 4 = 14)

6. Paint an L-shape made of a 4 by 2 rectangle and a 2 by 1 rectangle. What is its area?

10 square units (8 + 2)

Reflect

7. Reflect: How are area and perimeter different?

Area counts the unit squares inside; perimeter measures the distance around the outside.

Teacher notes

- Ask students to list every rectangle with area 12 and compare their perimeters (a long thin one has the biggest).
- Paint mode lets students build rectilinear figures for 3.MD.C.7d and check their own decomposition.
- Unknown side problems: give the perimeter and one side, then build it to check.

Standards: 3.MD.C.5, 3.MD.C.6, 3.MD.C.7, 3.MD.D.8. Accept any correct answer where several are possible, and ask students to explain with the tool's picture.